

MATUKU, Fiji

WMO: 916970

Lat: 19.133S

Lon: 179.750E

Elev: 16

StdP: 14.69

Time Zone: 12.00 (PCF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	66.2	67.7	53.5	61.0	70.7	55.3	65.1	71.1	21.4	72.3	19.0	74.1	3.4	160	0.272

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.4	91.3	84.8	89.5	83.2	87.5	81.0	85.4	90.5	83.9	89.0	81.8	87.1	8.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.0	178.2	89.9	82.4	169.1	89.1	80.1	156.4	86.3	49.8	90.4	48.1	89.0	45.6	87.1	88.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	15.7	13.7		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.8	81.3	81.8	81.8	79.9	77.5	75.1	73.4	73.6	74.7	76.0	78.4	80.0
	DBStd	4.00	2.50	2.58	2.95	2.20	2.19	2.22	2.23	2.68	3.00	2.82	2.84	2.80
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10141	972	890	986	897	853	754	726	732	740	807	853	931
	CDD65	4666	507	470	521	447	388	304	261	267	290	342	403	466
	CDH74	36337	5442	5214	5734	4170	2664	1223	605	750	1168	1812	3144	4411
	CDH80	7683	1532	1621	1764	731	180	25	17	13	92	190	544	974
(14) Wind	WSAvg	5.8	5.7	5.1	5.5	6.3	5.8	5.5	5.2	6.0	6.0	6.6	6.3	5.8
(15) Precipitation	PrecAvg	83.2	11.3	9.6	10.6	9.1	6.3	4.6	3.7	3.7	4.5	5.5	6.3	8.2
	PrecMax	116.2	26.0	16.7	16.4	16.1	14.2	11.4	6.4	9.7	8.6	10.4	16.0	15.5
	PrecMin	56.1	2.2	2.5	4.8	2.7	1.8	1.5	0.9	0.9	0.7	1.4	2.0	2.8
	PrecStd	15.2	5.6	3.6	3.2	3.9	3.2	2.5	1.5	2.1	2.2	2.4	3.1	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	92.4	93.4	87.6	84.1	81.9	81.9	81.6	83.7	85.4	89.0	90.5
		MCWB	84.2	85.0	86.0	79.9	77.2	75.9	75.1	74.5	76.0	77.5	82.7	82.9
	2%	DB	89.8	90.3	92.0	85.9	82.8	80.7	79.4	80.1	82.5	83.7	87.1	89.1
		MCWB	83.4	83.7	85.8	78.2	76.0	74.5	73.2	73.7	75.6	76.3	81.0	82.8
	5%	DB	87.5	88.1	89.5	84.7	81.9	79.7	78.1	78.9	80.9	82.2	85.2	87.2
		MCWB	81.3	81.5	84.0	77.5	75.2	73.6	72.1	72.9	74.6	75.1	78.4	80.8
	10%	DB	85.7	86.3	86.3	83.5	81.0	78.6	77.0	77.6	79.2	80.5	83.1	84.5
		MCWB	78.8	79.4	79.1	76.7	74.5	72.7	71.1	71.5	72.6	73.6	75.9	77.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	85.4	86.1	87.1	81.4	79.6	78.4	77.5	76.4	77.6	78.8	83.8	84.6
		MCDB	89.5	91.0	91.9	86.0	82.2	80.5	80.8	79.6	81.8	83.5	87.8	89.1
	2%	WB	84.3	84.4	86.0	80.0	78.1	76.4	75.2	74.9	76.6	77.2	82.1	83.4
		MCDB	89.1	89.5	91.6	84.1	81.5	79.2	77.7	78.9	80.9	82.5	86.7	88.7
	5%	WB	82.2	82.2	84.2	79.0	76.9	75.1	73.8	73.8	75.4	75.8	79.1	80.9
		MCDB	87.9	87.9	90.2	82.8	80.4	78.3	76.8	77.6	79.9	80.9	83.6	86.5
	10%	WB	79.6	80.4	80.3	78.0	75.8	74.0	72.4	72.8	73.9	74.8	77.2	78.6
		MCDB	84.4	85.4	84.9	82.0	79.9	77.7	76.0	76.8	78.3	79.6	81.8	83.0
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.4	6.9	6.2	6.2	5.7	5.9	5.9	6.0	6.2	6.4	6.5
		MCDBR	7.9	8.5	6.3	7.6	6.8	6.5	7.4	6.3	6.5	6.9	7.6	7.3
		MCWBR	3.8	4.1	3.3	4.0	3.8	3.8	4.3	3.7	4.0	3.7	4.0	3.6
	5% WB	MCDBR	6.4	7.6	4.5	7.6	6.2	5.8	6.8	6.1	5.5	5.9	6.4	6.2
		MCWBR	3.5	3.7	2.4	4.3	3.9	3.9	5.0	4.1	3.7	3.6	3.7	3.4
(40) Clear-Sky Solar Irradiance	taub	0.384	0.380	0.378	0.377	0.372	0.371	0.357	0.362	0.377	0.381	0.379	0.384	
	taud	2.634	2.653	2.671	2.654	2.629	2.618	2.646	2.637	2.582	2.583	2.619	2.629	
	Ebn at Noon	305	302	296	284	274	268	276	285	292	298	304	305	
	Edh at Noon	32	31	30	28	27	26	26	28	32	33	32	32	
(44) All-Sky Solar Radiation	RadAvg	2078	1959	1767	1496	1304	1153	1234	1400	1649	1851	2005	2052	
	RadStd	188	145	110	128	94	84	60	141	158	145	166	102	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	+0.45	N/A	N/A	+153	+153

Nomenclature: See separate page